→ Pseudocode for Genetic Algorithm for Optimization Problem

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Algorithm : Genetic Algorithm
Input : Pop-size , crossover-rate , mutation-rate , max-gen
Output : 'best-solution'
1. Define fitness (individual)
2. population ← random individuals (pop-size)
3. For generation 1 To max-gen do
      Evaluate fitness of each individual
      Update best solution if a better solution is found
      new-population ← empty

      while size (new-population) < pop-size do
            parent1 , parent2 ← Select (population)
            if random () < crossover rate then
                  child1, child2 ← crossover (parent1, parent2)
            else
                  child1, child2 ← parent1 , parent2

            Mutate child1, child2 with mutation-rate
            Add child1 , child2 to new population

      population ← new-population

4. return best-solution
```

Cyber Attack Detection Optimization

Select features and tune model parameters for network intrusion detection

Obj: Improve detection performance by reducing false alarms and cost computation.

Algorithm: Cyber_Attack_Detection_Optimization
Input: Pop_Size, max_gen, crossover_rate, mutation_rate, Data
Output: best_features, best_params

Pop ← init Population (Pop_Size)
for generation ← 0 to max_gen - 1 do
  for each individual in Pop do
    Feat ← Extract Features (individual.bits)
    Model ← Train (DATA.Train, Feat, param)
    Evaluate fitness
  next_pop = {}
  while size(next_pop) < pop size
    Parent1, parent2 ← select (pop)
    if (random() < crossover_rate then
      C1, C2 ← Crossover (parent1, parent2)
    else
      C1, C2 ← parent1, parent2
    Mutate C1, C2 with mutation_rate
    Add C1, C2 to new_pop

  pop = new_pop
  gen = gen + 1

BEST ← getBest (pop)
return BEST.bit, BEST.values

Using Genetic Algorithm inspired by natural evolution improves cyber security and intrusion detection system by automating feature selection and tuning the model by optimizing using best-parameter available

OUTPUT

Gen	Best Fitness	Features Selected	n_estimators	max_depth
1	0.7677	18	30	18
2	0.7762	14	90	21
3	0.7003	16	90	12
4	0.7717	18	150	9
5	0.7590	18	30	15
6	0.7544	16	110	9
7	0.7584	15	30	9
8	0.7777	17	150	15
9	0.7791	13	150	15
10	0.7842	18	70	24

CYBER ATTACK DETECTION OPTIMIZATION COMPLETE

Optimal Fitness Score : 0.7842

Optimized HyperParameters : n_estimator : 70, max_depth = 24

Selected Cyber Features (18/30) : ['duration', 'src_bytes', 'land', 'urgent', 'num_failed_logins'. . . . . 'dst_host_name, src_port_rate'].

O/P Seen Shelke 09/10/25